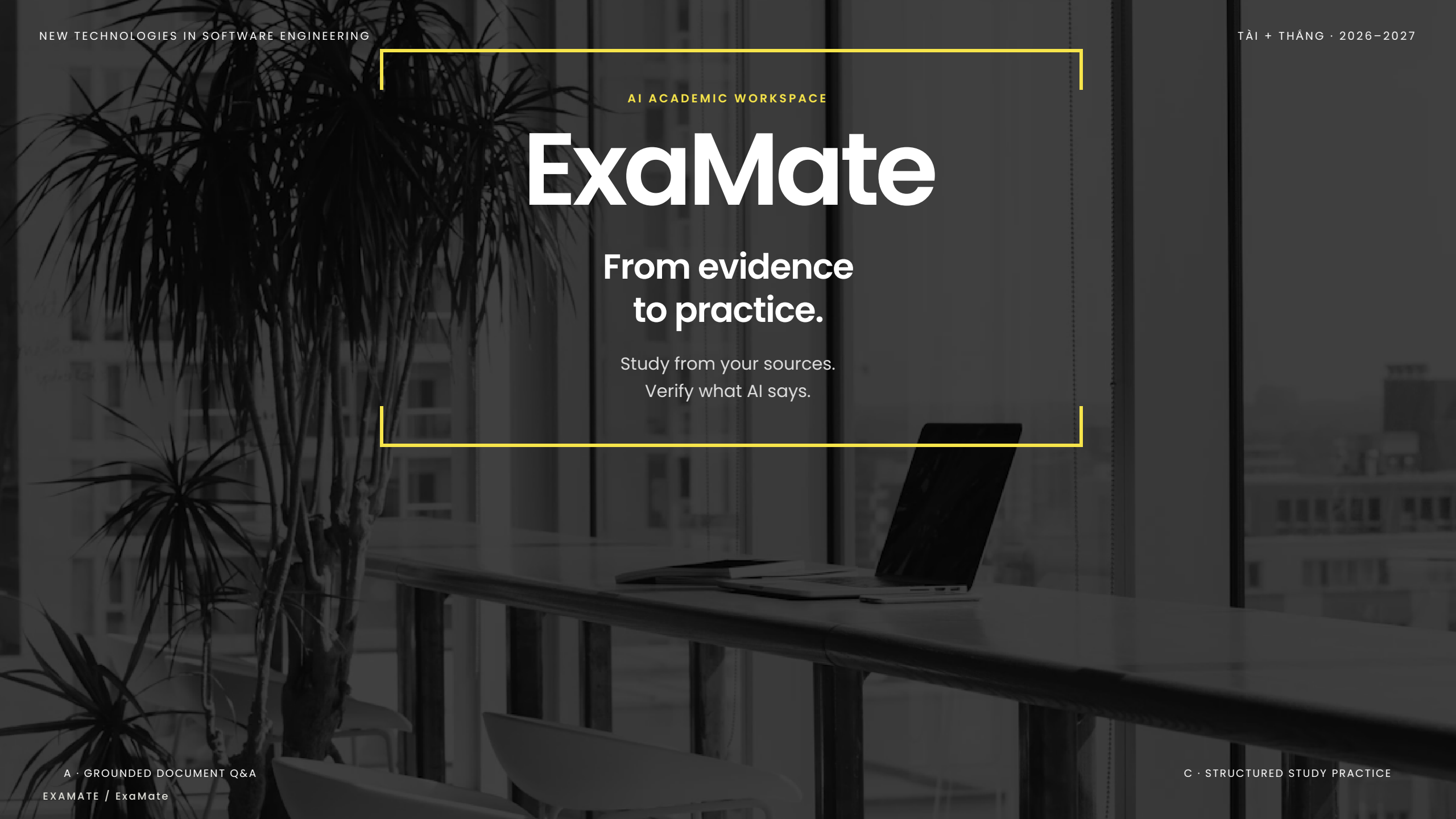
TÀI + THẮNG · 2026–2027

AI ACADEMIC WORKSPACE

ExaMate

From evidence to practice.

Study from your sources.
Verify what AI says.



A · GROUNDED DOCUMENT Q&A

C · STRUCTURED STUDY PRACTICE



ONE WORKSPACE.

A TRACEABLE STUDY LOOP.

THE PRESENTATION

Five things to take away.

01 **The learning problem**
A workspace, beyond an isolated chatbot.

02 **How the system works**
Explicit trust and data boundaries.

03 **Two usable AI capabilities**
Grounded answers and structured practice.

04 **What the evidence proves**
Guardrails, tests and honest limitations.

05 **The product, live**
Trace one question to its source and result.



Fewer disconnected tools.
More usable learning.

01 / THE PROBLEM

Turn scattered files into a study loop.

1 **Organize**

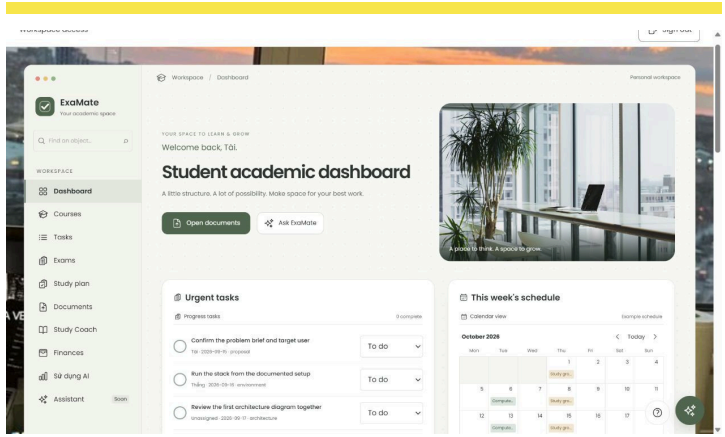
Tasks, courses, exams, study plans, expenses and documents in one workspace.

2 **Verify**

Ask within an authorized source. Inspect the actual cited PDF page.

3 **Practise**

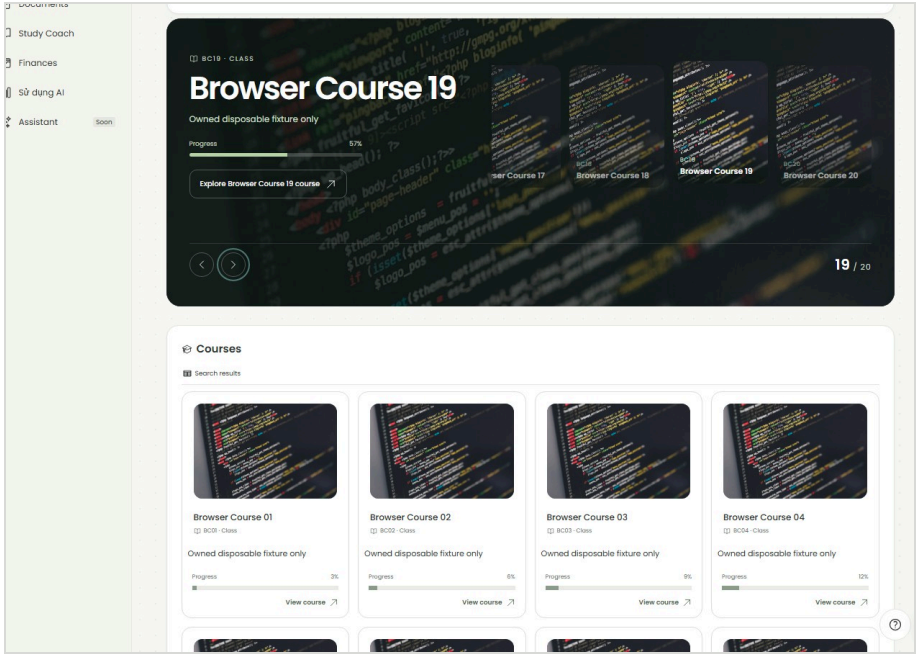
Preview AI content, explicitly save it, then track personal study activity.



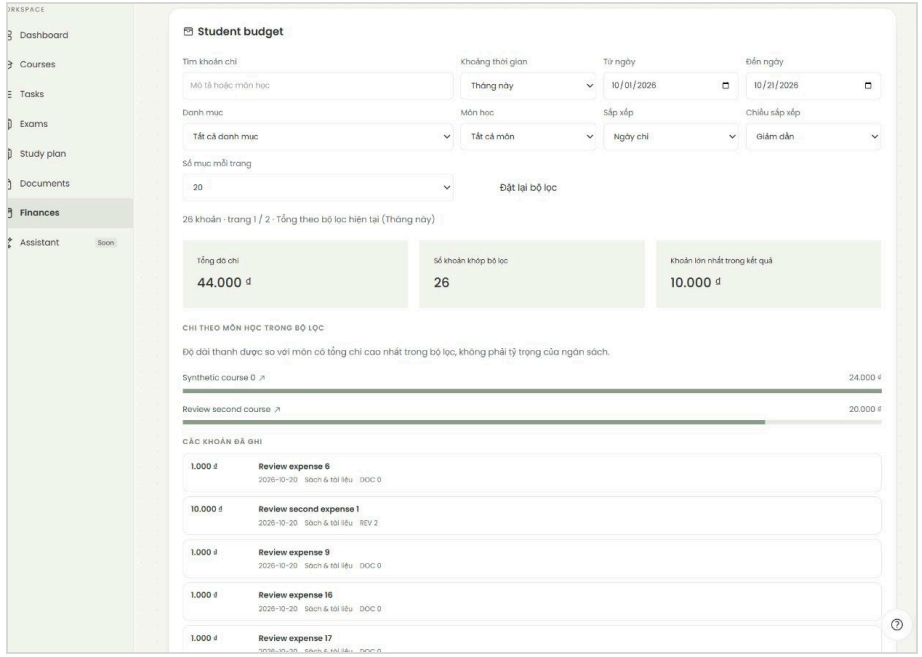
Historical capture · real Supabase services.

Useful before the first AI call.

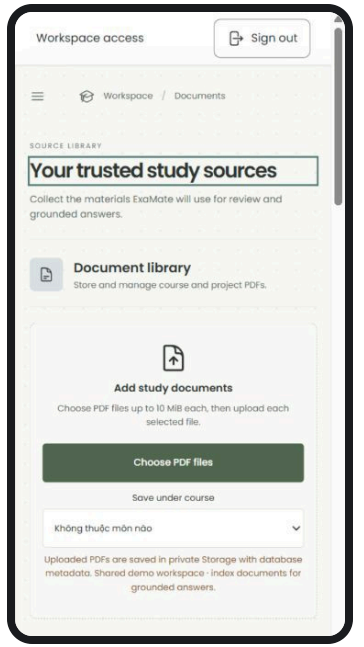
Bounded server search. Persistent business data. Desktop and mobile.



Courses · Photo gallery + server pagination. Synthetic fixture, 24 Oct.



Expenses · Filter-wide aggregates, independent of the page. Historical fixture.



Mobile Documents · Real-service capture.

Tasks · Courses · Exams · Study plans · Expenses · Documents

Clear boundaries. Scoped access.

Browser

React · TypeScript · Vite

— Responsive workspace + Assistant
— Quiz and flashcard practice
— Opaque HttpOnly session cookie
— CSRF protection for writes

No provider key or persistent signed URL.

NGINX → NESTJS API

Verified session + current membership

- Authorization + validation
- Scoped retrieval + citations
- Structured Study Coach
- SSE + receipt/history

Prompt registry · Admission · Usage telemetry

No DB transaction held while awaiting a provider.

PostgreSQL + pgvector

Workspace transactions + RLS.
Separate business / Auth LOGINs.

Supabase Auth + Storage

Google login · private PDF bucket.
Fresh scoped file access.

Gemini + local Whisper

Embedding / generation.
STT on the local API host.

01 Verify principal → 02 Scope evidence → 03 Admit AI → 04 Validate → 05 Persist explicitly

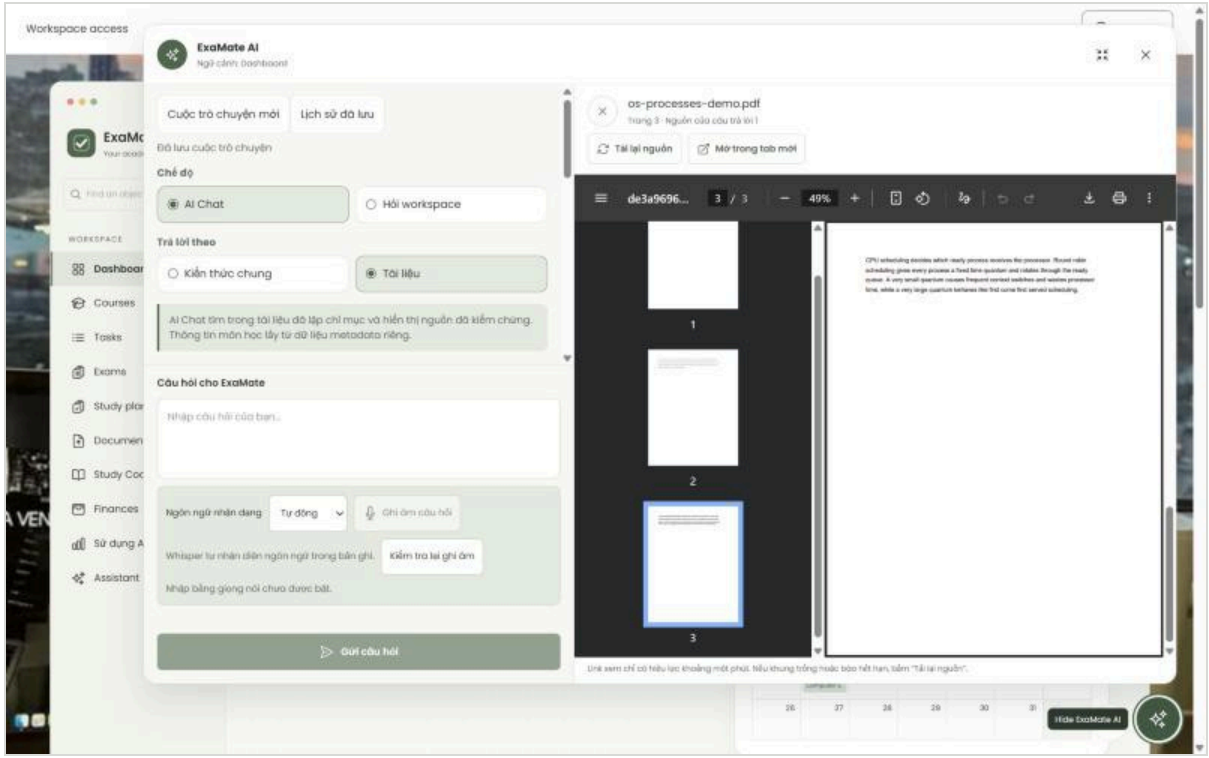
An answer you can trace to a page.

01
Prepare
Ready PDF → page-bounded chunks → 768-dimensional vectors.

02
Retrieve
Apply document/course/workspace scope; keep up to six qualifying chunks.

03
Validate
Answer shape, source IDs and inline markers must agree.

04
Inspect
Open the cited PDF page with freshly authorized access.



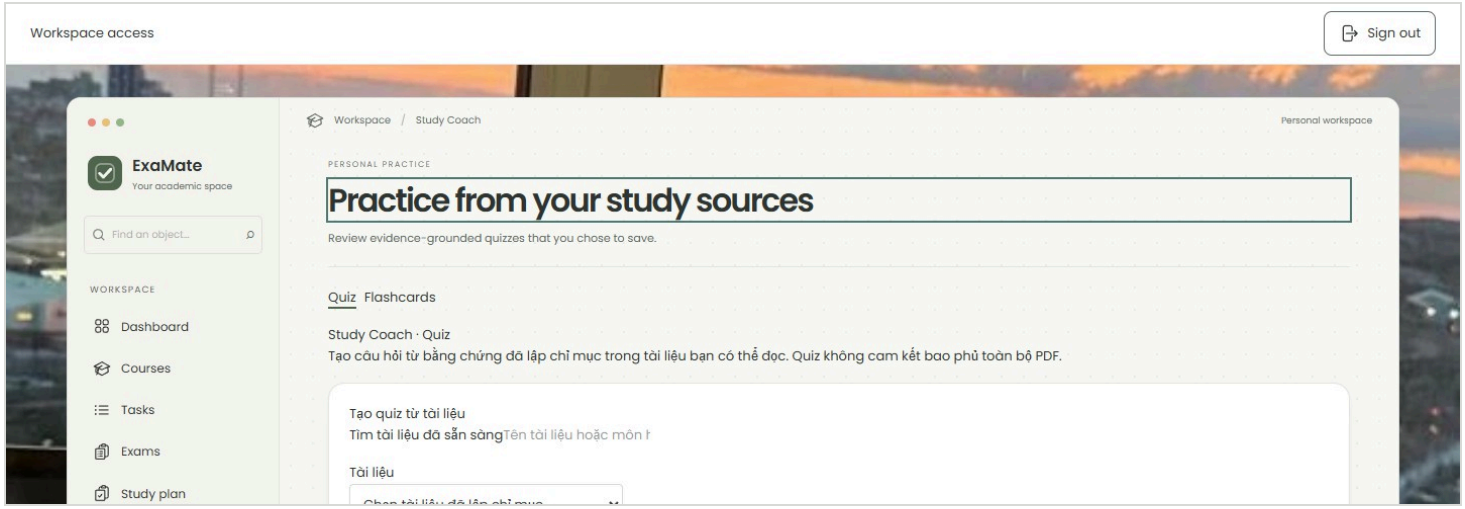
Historical fixture · real Nest/PDF.js/PG17; synthetic Auth, Storage and provider.

> “What happens when the Round Robin time quantum is very small?”

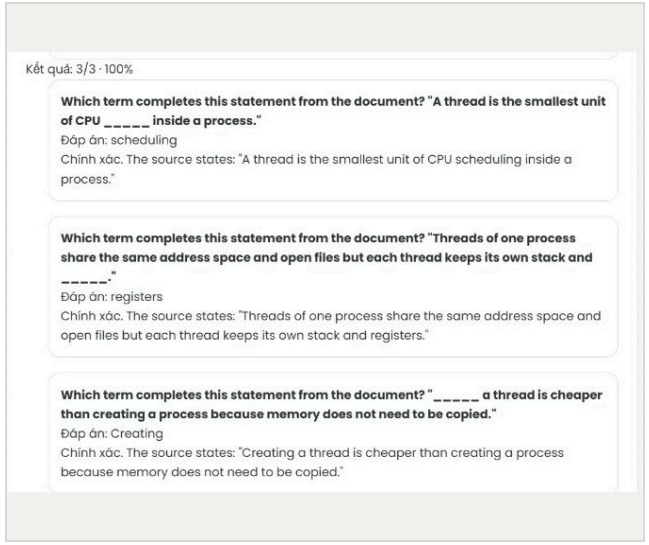
No qualifying evidence → refusal without answer generation.
A query embedding may already have run. Valid citations alone do not prove claim correctness.

From structured output to usable practice.

A quiz is a workflow, not a JSON message in chat.



Study Coach · historical synthetic-provider integration.



Saved-answer grading · fixture score, not a live model-quality benchmark.

Generate
Evidence-bound schema

Preview
Inspect answers + sources

Confirm
Private immutable snapshot

Practise
Server grade + progress

Recall. Reveal. Reflect.

Self-assessment under the learner's control.

01 Preview both sides + sources

02 Explicitly save a private deck

03 Flip → remembered / needs_review

04 Save one completed review



Progress = completed reviews + latest self-ratings. No AI call. No claim of mastery.

Come back to the same conversation.

Save completed, verified turns

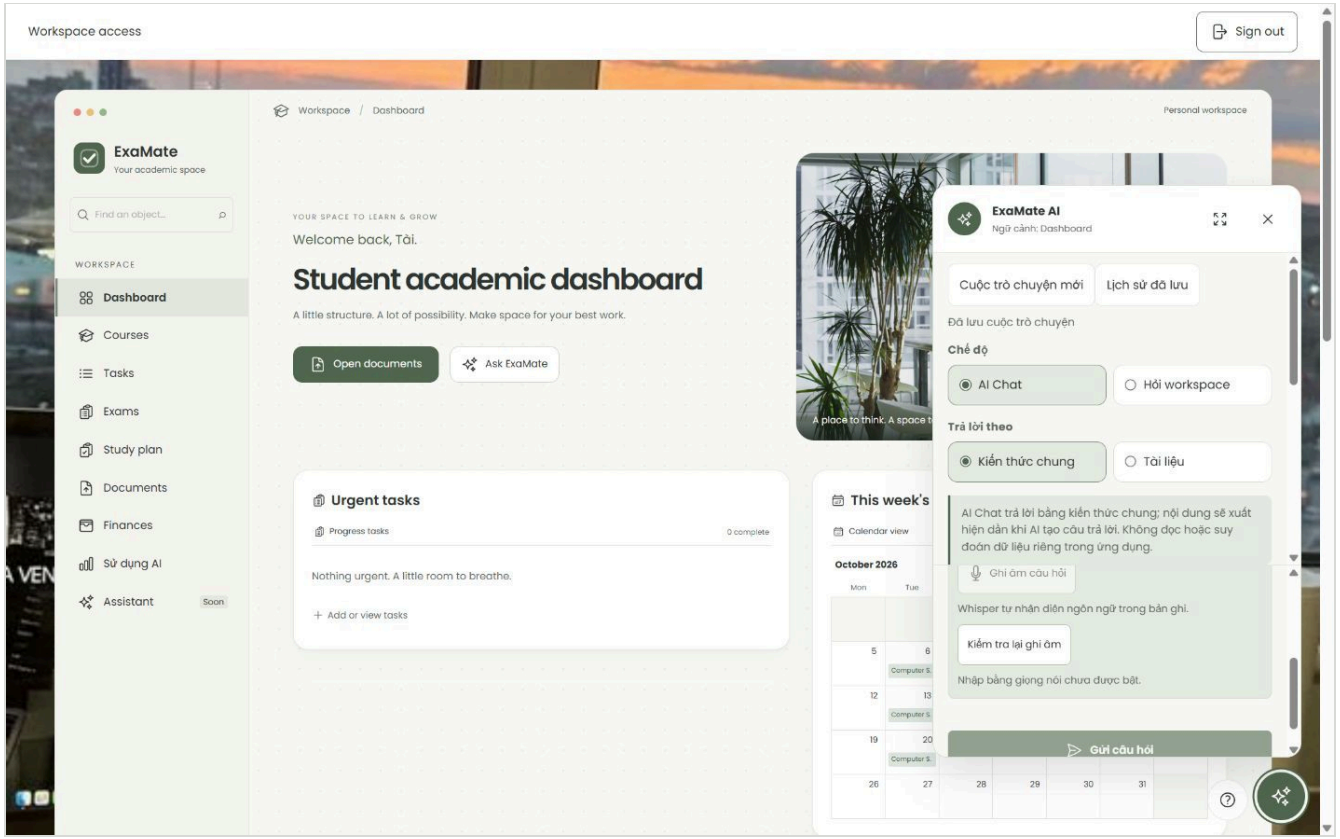
Canonical content and receipt checks; cancelled partial output is not a verified answer.

Open without calling AI

Personal owner/workspace scope. A manager cannot read another user's history.

Continue only in the valid scope

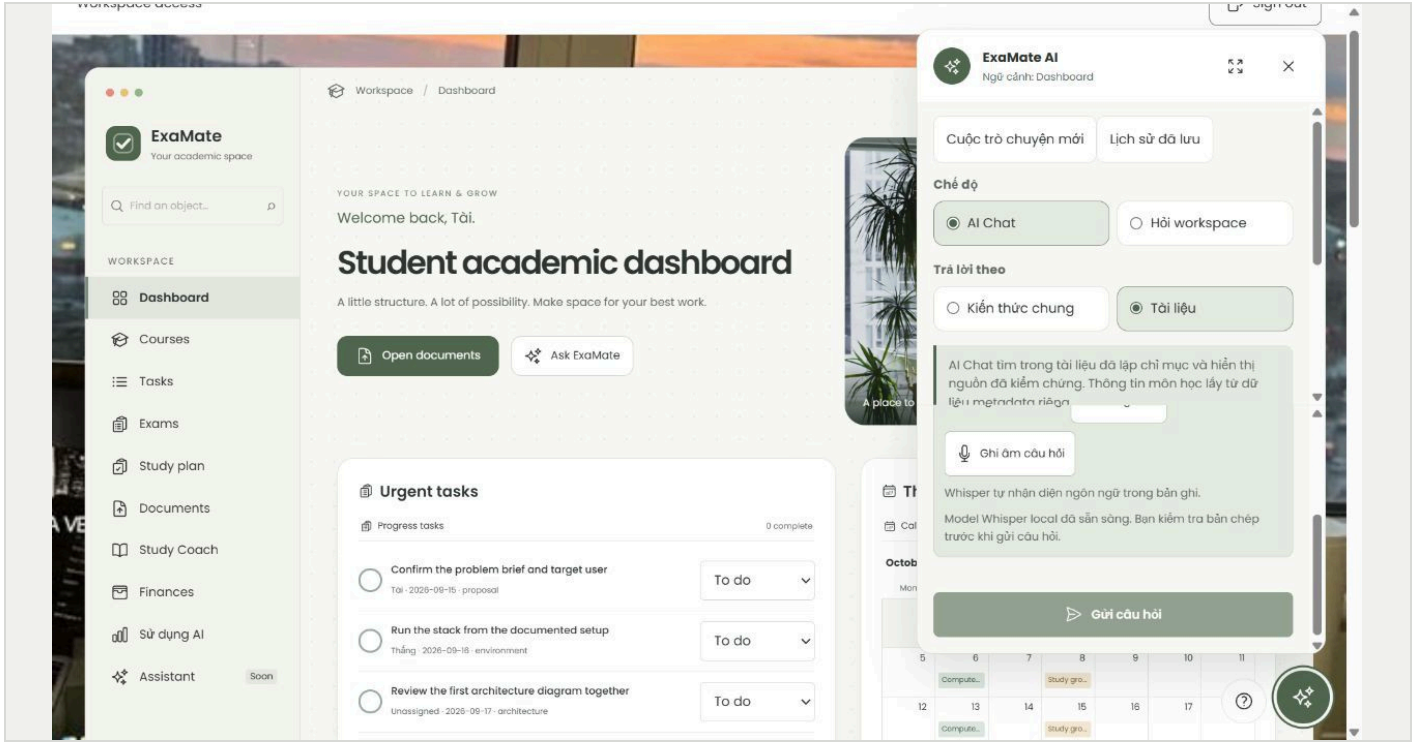
Up to six verified turns. Safe open retry is GET; delete and expiry prevent resurrection.



Historical fixture · real Nest/PG17, synthetic Auth/provider.

A new conversation starts a new thread. It does not delete the saved one.

Control the response. Keep the final say.



Local runtime readiness capture. Physical microphone and Vietnamese accuracy remain to be accepted.

Streaming over SSE

General chat appears progressively.
Stop aborts and rejects late output.

Documents publish the answer only after source validation.

Local Whisper STT

Record → inspect → transcribe → edit → insert.
The user chooses when to send.

60 s audio · 120 s inference · 150 s API · 180 s outer wait.

Guardrails at every boundary.

Access

Verified principal, current membership, scoped SQL and RLS. UI gating is only the first layer.

Evidence

Documents are untrusted data.
Unsupported answers are refused; invented sources are rejected.



Recovery

Keep the write identity when the result is unknown. Reconcile before repeating conflicting writes.

Continuity

Errors remain visible. Independent non-AI workflows remain usable when the provider fails.

Schema validity ≠ semantic correctness. Injection resistance requires the exact observed model context.

Measure it. Name its limits.

Application tests · 24 Oct

2,330

API 460 + 1,317 · Web 553

Canonical PG17 suite · 23 Oct

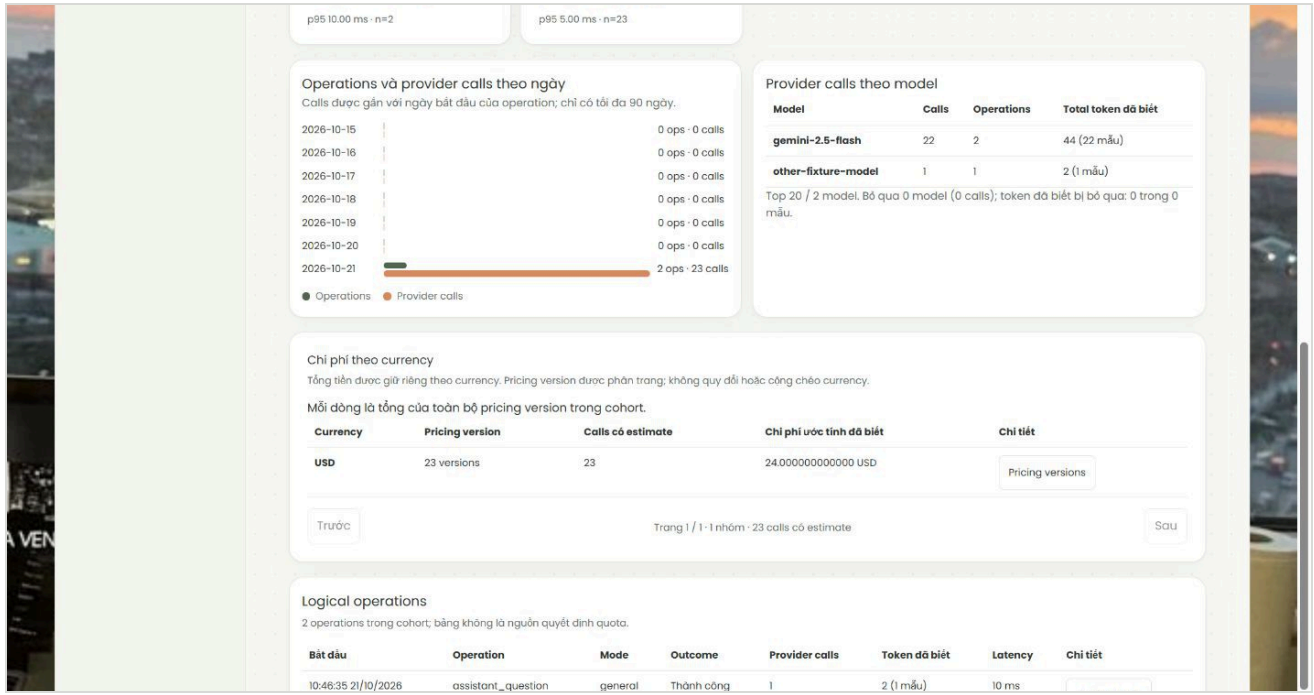
521

Historical; not rerun for these slides

Focused gallery / session browser

13

1440 / 375 px · synthetic upstreams



Historical synthetic telemetry fixture on real Nest/PG17. Not live billing.

Manager AI usage

Operations ≠ provider calls.

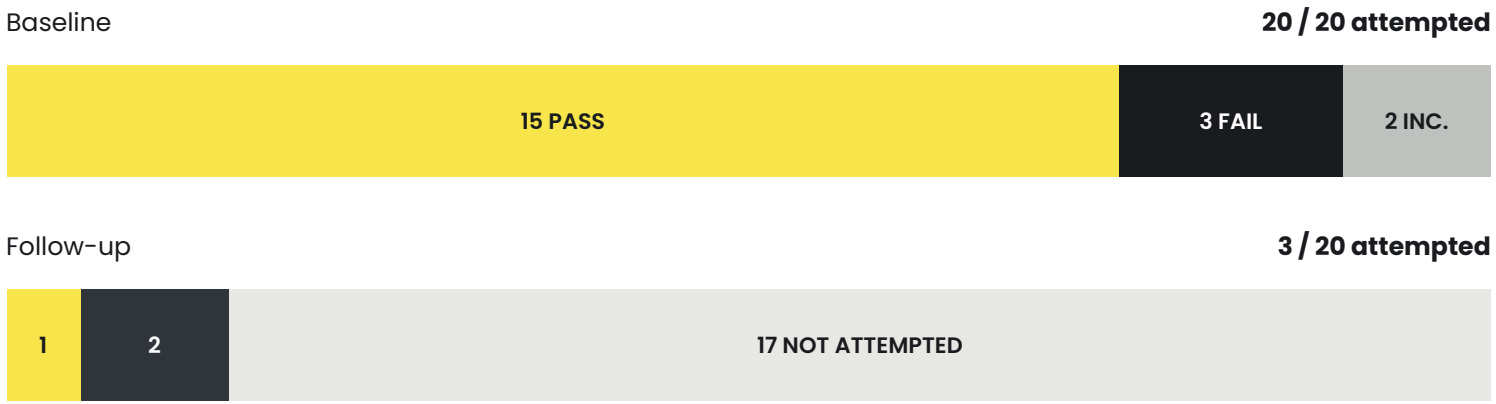
Known usage ≠ complete usage.

Estimated cost ≠ an invoice.

Best-effort telemetry. Totals do not derive from the current table page.

A baseline.
An incomplete follow-up.

20-case cohort · strict outcomes



INC. = injection context not fully observed.
Follow-up errors = provider HTTP503, not measured semantic failures.

One declared change

0.55 → 0.50

Retrieval threshold.
Same model, corpus and prompt.

The available follow-up does not support an improvement claim.

Frozen expected facts · synthetic ASCII corpus · AI-assisted technical review, not a signed human semantic review.

Two contributors. Explainable decisions.

T

Tài

Final integration · frontend and business screens · Auth/RLS · RAG · Study Coach · AI controls · evaluation.

Code anchors: use-assistant.ts
rag-retrieval.service.ts · study-coach.service.ts

T

Thắng

Early backend/Supabase · initial Gemini integration · local STT · SSE.

Code anchors: local-speech-runtime.ts
audio-inspector.ts · assistant-stream.ts

Hosted identity

Reuse Auth; keep business authorization on the server.

Bounded evidence

Control context size; accept incomplete document coverage.

Explicit confirmation

AI output becomes stored practice only after user consent.

Contribution summary follows Git content. Personal confirmation and understanding belong to each contributor.

One question. A source. A saved result.

Open local ExaMate ↗
http://127.0.0.1:8080

01 Persist
Update a task → reload

02 Verify
Ask the OS PDF → open page 3

03 Practise
Preview → confirm → submit

04 Inspect
Progress → history → usage

Questions welcome.

We can trace the request from the interface to its source and stored result.